Fiscal Year:	FY 2019	**Task Last Updated:**	FY 05/31/2019
PI Name:	Hasenstein, Karl Ph.D.		
Project Title:	Assessment of Nutritional Value and Growth Parameters of Space-Grown Plants		
Division Name:	Space Biology		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Plant Biology		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	(1) Bioregenerative Life Support		
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Zip Code:	70504-2451	**Congressional District:**	3
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	Space Biology Unsolicited
Start Date:	08/01/2017	**End Date:**	07/31/2021
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:	2	**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:	2	**Monitoring Center:**	NASA KSC
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Flight Assignment:	NOTE: End date changed to 7/31/2021 per NSSC information (Ed., 5/4/2020)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
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	The goal of the proposed work is to determine the effects of space flight on growth, performance (productivity and time to maturation), metabolism, and genetic changes in radish plants (Raphanus sativus). Plants will be grown in the ADVANCED PLANT HABITAT (APH) hardware on the International Space Station (ISS) and in comparable experiments in ground controls. The research conforms to NASA's top future goal of long-term space exploration and terraforming other planets and satellites. These goals depend on nutritionally valuable food crops that can be reliably grown in an environment that is not only defined by altered gravity or weightless environment but also different atmospheric conditions. In contrast to studying direct effects of gravity on plants, our approach focuses on secondary, derived conditions such as altered convection, transpiration, watering conditions, CO2, other gaseous components, and LED illumination, which creates a characteristic energy (heat and light) gradient within the growth volume.		

Task Description:	Radishes are good candidate plants to grow on the ISS because they mature quickly (harvest after = 28 days after planting), have been grown successfully on the ISS (Massa et al., 2013); genetic and metabolic information is readily available (Shen et al., 2013); and they are nutritionally desirable and provide a wealth of information that can thoroughly be analyzed by biochemical, physiological, and genetic approaches. Radishes are good sources of minerals (especially K, Ca, Fe, and Zn), complex carbohydrates, vitamin C, dietary fibers, folic acid (vitamin B9), anthocyanins, and glucosinolates, a class of secondary metabolites characteristic for Brassicaceae. In addition to their roots, certain varieties of radish are used for their leaves, siliques, and oil. The nutritional value and versatility of these vegetables makes them good candidates as a crop plant for astronauts. The small seed size (<4 mm), longevity of seeds (up to five years), small plant size, and the ability to consume the entire plant makes radishes ideally suited for space biology. Previous studies have focused on the effect of gravity and shown that space conditions allow successful cultivation of Brassica species (Musgrave et al., 2000). However, the quality, biomass, sensitivity of growth to day length, and (constant) temperature are important parameters for this species that have yet to be studied, especially in light of results that seed development in space cultures was compromised (Kuang et al., 2005), especially when oxygen was limited (Kuang et al., 1998). Equally unknown is the effect of environmental conditions in spacecrafts on the performance of plants. Understandably, the emphasis on space biology focuses on the effect of weightlessness and gravity on plant orientation and mechanical properties (Hoson 1999). However, conditions that are the consequence of weightlessness, e.g., lack of convection or altered atmosphere, are equally significant but unknown. Our proposal intends to measure the metabolic, physiological, and genetic responses of radishes grown on the ISS and identify the effect of the space environment on metabolite accumulation, flavor, enzyme activity, mineral uptake, and generation time. This knowledge will be essential for the transition from Earth-bound cultivation of plants to growth conditions in space; the research is important for the establishment of human colonies on the Moon, Mars, and farther outposts. References Hoson T (1999) Interaction of gravity with other environmental factors in growth and development: An introduction. In: Duke PJ, Barlow PW, Hoson T, Mogami Y (eds) Life Sciences: Microgravity Research I, pp 1971-1974 Kuang A, Crispi M, Musgrave ME (1998) Control of seed development in Arabidopsis thaliana by atmospheric oxygen. Plant Cell and Environment 21: 71-78 Kuang A, Popova A, McClure G, Musgrave ME (2005) Dynamics of storage reserve deposition during Brassica rapa L. pollen and seed development in microgravity. Int J Plant Sciences 166: 85-96 Massa G, Newsham G, Hummerick ME, Caro JL, Stutte GW, Morrow RC, Wheeler RM (2013) Preliminary Species and Media Selection for the VEGGIE Space Hardware. Grav Space Res 1: 95-106 Musgrave M, Kuang A, Xiao Y, Stout S, Bingham G, Briarty L, Levinskikh M, Sychev V, Podolski I (2000) Gravity independence of seed-to-seed cycling in Brassica rapa. Planta 210: 400-406 Shen D, Sun H, Huang M, Zheng Y, Li X, Fei Z (2013) RadishBase: A Database for Genomics and Genetics of Radish. Plant Cell Physiol 54: E3-+
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The objective to grow plants under space conditions requires optimization of auxiliary parameters such as light intensity, quality, watering, substrate definition, fertilization that all have immediate implications for more efficient agricultural practices. The data that we will obtain examine much more than simple yield (mass) of the plants but include the mineral content, propensity to flower, flavor components, and water use. All attempts will not only optimize plant culture under elevated CO2, limited water, and controlled fertilizer provision, but also investigate the genetic responses of plants to mitigate space (or agricultural) stress. The overall result of the project will therefore benefit our generic understanding of factors that control plant growth and productivity in space and on Earth.
	Choice of seed material: We tested germination rate and percentage of germination of nine different varieties of Raphanus sativus from two sources and determined that the variety Cherry Belle, which completes the growth cycle in approximately 23 days is best suited for all future experiments. Seed sanitation: To accommodate long-term storage and minimize surface-borne microorganisms, we sanitized seeds based on several protocols, including bleach and ethanol. The high mortality rate after as little as one-minute exposure to 70% ethanol resulted in the following protocol: Substrate development. The predetermined use of arcillite (calcined Montmorillonite) for prior Kennedy Space Center (KSC)-based experiments, its porous structure (defined for 1 to 3 mm grains), inert properties at neutral pH, low density (~0.69 g×cm-3), and hydrophilicity suggested its use as growth substrate. The initial strategy relied on the use of slow-release fertilizer (Nutricote) as plant nutrient resource. However, the slow release given the fast maturation required a better and more uniform provision of mineral elements. The chosen strategy now relies on inclusion of a modified Murashige and Skoog medium. Because of corrosion concerns, this medium is now prepared without chloride. While arcillite without fertilizer is inadequate for growth a single application of MS nutrients is sufficient for at least two crops (12 plants per 2 L volume of arcillite). Plant growth was superior to Nutricote. Initial tests were performed in common plastic containers; later experiments were based on science carrier replicates (3D-printed boxes and covers of the same size as the science carrier). These containers enhanced the fidelity of the experiments by providing the same substrate depth and volume for root growth. Because water content is critical for the formation of radishes, we studied the water consumption. Based on five 4-week long experiments, the water consumption turned out a function of leaf area, light intensity, and temperature. The water loss increases with age (produced leaf material) from an initial range of 80 to 100 mL per day per 12 plants to 500-600 mL per day as the plants mature. The low water capacity of arcillite requires daily watering, especially in the last two weeks of growth. Light conditions: Initial experiments relied on a regular growth chamber with high intensity white fluorescent lights (Percival E-54U). The PAR light intensity of this chamber is about 130 µmol m-2 s-1. A newly acquired high intensity

Task Progress:	LED based light fixture (Heliospectra RX30) has programmable spectral light qualities up to 1000 µmol m-2 s-1. The increased light intensity increased water consumption but produced considerably larger radishes and plants. Analytical work: We examined the mineral composition of harvested plant tissue (leaf and radish), and the ion activity of arcillite substrate. The mineral composition of Cherry Belle radishes confirms the high amount of potassium in all tissues. There are no substantial differences between mineral content of leaves and radish tissue on a dry weight basis. Part of the flight experiment aims to assess the effect of space conditions on the secondary compounds. Brassicaceae are known as a rich source of glucosinolates. Therefore, we examined the yield of glucosinolate contents in fresh and stored (frozen) material. The average yield was 120 µmol glucosinolates per 100 mg fresh weight. This amount was not affected by frozen storage up to three months. These data are important because flight material will only be analyzable after extended frozen storage on station. In contrast to metabolites, the longevity of some mRNA was reduced the longer tissue was stored frozen. Among the tested transcription products (mRNAs) peroxidase precursor (EU369394.1) showed a continuous decline in bulb (= radish) tissue but this decline was not observed in leaf tissue. Leaf tissue also contained significantly less transcript compared to bulb tissue. The data indicate that we can reliably quantify gene transcription, but they also indicate that culture conditions, light quality, fertilizer status and storage conditions will affect the outcome of the analyses. The nitrogen content of tissue depends on light and photoperiod and fertilizer content. Since a lot of effort went into analyzing corrosion susceptibility of the AHP hardware, we tested the amount of nitrogen in tissue when plants were grown in Murashige-Skoog vs. Hoagland medium. The overall nitrogen quantity in leaf and bulb tissue of MS medium-grown plants was about twice that of Hoagland-grown plants. In addition to nutrients, the quality of light also affected growth. We discovered that culture conditions with high intensity of blue light reduced water consumption and enhanced biomass compared to red or mixed (red & blue) light conditions without affecting the overall mineral content. The work performed in the first 18 months was necessary for the preparation of the Science Verification Test (SVT), which started May 16, 2019. The SVT compared the growth of radish in two configurations – either five or nine plants per science carrier quadrant. This test will evaluate which number of plants can be supported in a given area under otherwise identical conditions. Illumination and therefore photosynthesis will be reduced when plants grow too densely. Therefore, the SVT will not only provide valuable information on hardware performance but also inform our experimental evaluation of growth parameters.
Bibliography Type:	Description: (Last Updated: 01/20/2026)
Abstracts for Journals and Proceedings	Pitrucha B, John SP, Abou-Issa F, Hasenstein KH. "Arcillite as Substrate for Plant Growth in Microgravity." 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. Program and abstracts. 34th Annual Meeting of the American Society for Gravitational and Space Research, Bethesda, MD, October 31-November 3, 2018. , Oct-2018
Articles in Peer-reviewed Journals	Ajala C, Hasenstein KH. "Augmentation of root gravitropism by hypocotyl curvature in Brassica rapa seedlings." Plant Science. 2019 Aug;285:214-23. Available online 25 May 2019. https://doi.org/10.1016/j.plantsci.2019.05.017 ; PMID: 31203886 , Aug-2019